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Week 6: Image Processing

Reading: Week 6 Lecture (30 min)

Ungraded Pugin: 6.1 Overview of Image Processing (4 min)

Ungraded Pugin: 6.2 Fourier Transforms (17 min)

Practice Quiz: 6.2 Fourier Transform Self-check Quiz (Start)

Reading: 6.3 Video (10 min)

Ungraded Pugin: 6.3 Convolution Theorem (8 min)

Practice Quiz: 6.3 Convolution Theorem Self-check Quiz (Start)

Ungraded Pugin: 6.4 Image Filtering in Frequency Domain (14 min)

Practice Quiz: 6.4 Image Filtering in Frequency Domain (5 min)

Ungraded Pugin: 6.5 Deconvolution (12 min)

Practice Quiz: 6.5 Deconvolution Self-check Quiz (5 min)

Reading: 6.6 Video (10 min)

Ungraded Pugin: 6.6 Sampling Theory and Aliasing (13 min)

Practice Quiz: 6.6 Sampling Theory and Aliasing Self-check Quiz (10 min)

Discussion Prompt: Work in Questions and Feedback

Week 6 Quiz

Post-Course Survey

● Congratulations! You passed!

Grade received 100% (To pass 50% or higher)

Go to next item

6.3 Convolution Theorem Self-check Quiz

1. $f(x)$ = Square wave with frequency $1\text{ kHz} \leq f(x) \leq \sin(\omega x)$ where $\omega = 2\pi H\text{ s}$.
 $g(x) = f(x) * h(x)$ where $h(x) = \text{rect}(x/2)$.
Hint: Recall the Fourier transform of a square wave and the Fourier transform of a rect function.
☐ Combination of 2 sinusoids of 1 kHz and 2 kHz
☒ Single sinusoid with frequency 1 kHz and phase shifted by π
☐ Single sinusoid with frequency 1 kHz phase shifted by $\frac{\pi}{2}$
☐ Single sinusoid with frequency 1 kHz phase shifted by π
☒ Correct
Square wave has frequency $1\text{ kHz} \leq f(x) \leq 3\text{ kHz}$ and so on. $H(f)$ will be a box function stretching from -2 kHz to 2 kHz . $F(u)H(u)$ will just have a single 1 kHz component.

1 / 1 point

Try again

2. Given two signals $f(t) = \cos(t)$ and $g(t) = \sin(t)$. Calculate the signal $h(t)$ which is constructed using convolving $f(t)$ and $g(t)$ i.e. $h(t) = f(t) * g(t)$.
Note: $\cos(t)$ is the unit step function
☐ $\frac{1}{2}\cos(t)$
☐ $\frac{1}{2}\sin(t)$
☐ $\frac{1}{2}\cos(t)$
☒ $\frac{1}{2}\sin(t)$
☐ Correct
Calculate the convolution empirically

1 / 1 point

Report as issue